

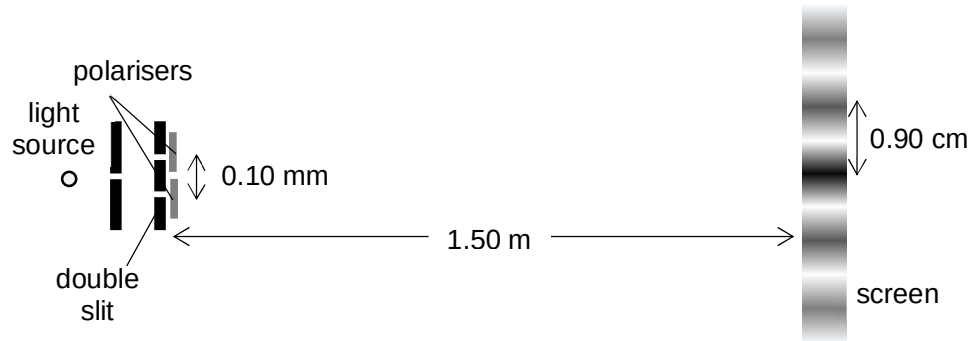
- 4 (a) Explain what is meant by *polarised* light. [1]

Light in which the oscillations of the electromagnetic fields are all in a single plane

- (b) Explain why two coherent sources of light that are polarised in planes perpendicular to each other will not produce interference fringes. [1]

The displacements due to the two waves are in perpendicular axes, thus their vector sum will not be able to produce maxima and minima, since they can never be in the same or in opposite directions.

- (c) Fig. 4.1 shows the setup for a Young's double slit experiment. The polarisers are initially aligned such that the light from the slits are polarised in parallel planes. The slit separation of the double slit is 0.10 mm, the distance between the double slit and the screen is 1.50 m and the separation of the fringes produced is 0.90 cm.

**Fig. 4.1**

- (i) Calculate the wavelength of the light source. [2]

$$\lambda = a \times D = 0.10 \times 10^{-3} \times 0.90 \times 10^{-2} / 1.50 = 6.00 \times 10^{-7} \text{ m}$$

- (ii) One of the polarisers is rotated by an angle of 45° about an axis parallel to the incident light. Describe the appearance of the fringes with reference to its original appearance. [2]

Only the components of the light waves that are oscillating in parallel planes will interfere to form maxima and minima. The components that are oscillating in perpendicular planes will not be affected by each other. Thus the maxima will be less bright and the minima will be brighter.